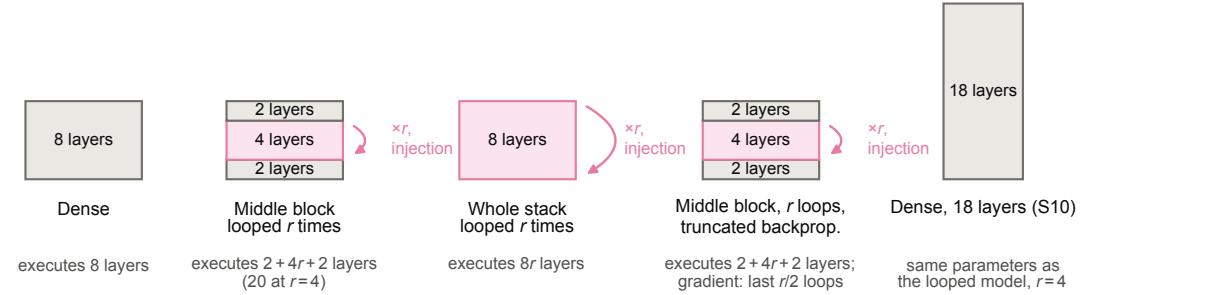
(a) The language models: the dense reference, three kinds of loop and the dense control of S10
8 layers
Dense
executes 8 layers
2 layers
4 layers
2 layers
Middle block looped r times
executes 2 + 4r + 2 layers (20 at r = 4)
×r, injection
8 layers
Whole stack looped r times
executes 8r layers
×r, injection
2 layers
4 layers
2 layers
Middle block, r loops, truncated backprop.
executes 2 + 4r + 2 layers; gradient: last r/2 loops
×r, injection
18 layers
Dense, 18 layers (S10)
same parameters as the looped model, r = 4
(b) The chain task: one problem with d = 3 steps, and how each model answers it
prompt
BOS
25
;
LUT
7
;
LUT
3
;
LUT
19
;
=>
3d + 4 = 13 tokens; x1 = T7(25), x2 = T3(x1), answer = T19(x2)
Looped, answer only
prompt
core run r times, here r = d: loop 1 → x1, loop 2 → x2, loop 3 → answer
answer
more loops at inference → longer chains
Chain of thought (dense)
prompt
8 or 36 layers
x1
x2
answer
writes the chain out: d tokens, the last of which is the answer
Pause tokens (dense)
prompt
pause
pause
pause
8 layers
answer
d pause tokens before the answer
Dense, equal depth
prompt
20 or 36 layers, once
answer
the executed depth of 4 or 8 loops, 2.4--4.4× the parameters